

1 Solution — GIAM 1.6

Problem 1

1.1 Problem

Rational Approximation is a field of mathematics that has received much study. The main idea is to find rational numbers that are very good approximations to given irrationals. For example, $22/7$ is a well-known rational approximation to π . Find good rational approximations to $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$, and e .

1.2 The Method — Continued Fractions

There is one systematic way to produce the *best possible* rational approximations, and it uses exactly the machinery of Section 1.5. Run the Euclidean algorithm on the number (allowing an irrational input): the successive quotients form a **continued fraction**, and truncating it gives the **convergents** p/q — the fractions that approximate better than *any* fraction with a smaller denominator. The famous $22/7 \approx \pi$ is just the first interesting convergent of π .

Below, " $[a_0; a_1, a_2, \dots]$ " denotes $a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \dots}}$, and a bar marks the repeating block.

1.3 The Answers

number	good approximation	decimal	error
$\sqrt{2}$	$\boxed{\frac{577}{408}}$	$1.4142156\dots$	$\approx +2 \times 10^{-6}$
$\sqrt{3}$	$\boxed{\frac{97}{56}}$	$1.7321429\dots$	$\approx +9 \times 10^{-5}$

$\sqrt{5}$	$\boxed{\frac{682}{305}}$	$2.2360656\dots \approx -2 \times 10^{-6}$
e	$\boxed{\frac{193}{71}}$	$2.7183099\dots \approx +3 \times 10^{-5}$

Table 1.1

Simpler (slightly less accurate) choices are $\sqrt{2} \approx \frac{99}{70}$, $\sqrt{3} \approx \frac{71}{41}$, $\sqrt{5} \approx \frac{161}{72}$, $e \approx \frac{87}{32}$.

1.4 Working — The Four Continued Fractions

$\sqrt{2} = [1; \overline{2}] = [1; 2, 2, \dots]$. Convergents:

$$1, \frac{3}{2}, \frac{7}{5}, \frac{17}{12}, \frac{41}{29}, \frac{99}{70}, \frac{239}{169}, \frac{577}{408}, \dots$$

$\sqrt{3} = [1; \overline{1, 2}] = [1; 1, 2, 1, 2, \dots]$. Convergents:

$$1, 2, \frac{5}{3}, \frac{7}{4}, \frac{19}{11}, \frac{26}{15}, \frac{71}{41}, \frac{97}{56}, \dots$$

$\sqrt{5} = [2; \overline{4}] = [2; 4, 4, 4, \dots]$. Convergents:

$$2, \frac{9}{4}, \frac{38}{17}, \frac{161}{72}, \frac{682}{305}, \dots$$

$e = [2; 1, 2, 1, 1, 4, 1, 1, 6, 1, 1, 8, \dots]$ (the pattern $1, 2k, 1$ repeats; not periodic).
Convergents:

$$2, 3, \frac{8}{3}, \frac{11}{4}, \frac{19}{7}, \frac{87}{32}, \frac{106}{39}, \frac{193}{71}, \dots$$

1.5 The Convergents at a Glance

Run the Euclidean algorithm on the irrational: its quotients give a continued fraction, whose truncations (convergents) are the best rational approximations. Each one alternates over / under the true value.

$\sqrt{2} = 1.41421356\dots$ continued fraction $[1; 2, 2, 2, 2, \dots]$ (period 2) convergents: 1, 3/2, 7/5, 17/12, 41/29, 99/70 , 577/408 $99/70 = 1.4142857\dots$ (error $\approx +7 \times 10^{-5}$) $577/408 = 1.41421569\dots$ (error $\approx +2 \times 10^{-6}$)	$\sqrt{2} \approx 577/408$
$\sqrt{3} = 1.73205081\dots$ continued fraction $[1; 1, 2, 1, 2, \dots]$ (period 1,2) convergents: 1, 2, 5/3, 7/4, 19/11, 26/15, 71/41, 97/56 $71/41 = 1.7317073\dots$ (error $\approx -3 \times 10^{-4}$) $97/56 = 1.7321429\dots$ (error $\approx +9 \times 10^{-5}$)	$\sqrt{3} \approx 97/56$
$\sqrt{5} = 2.23606798\dots$ continued fraction $[2; 4, 4, 4, 4, \dots]$ (period 4) convergents: 2, 9/4, 38/17, 161/72 , 682/305 $161/72 = 2.2361111\dots$ (error $\approx +4 \times 10^{-5}$) $682/305 = 2.23606557\dots$ (error $\approx -2 \times 10^{-6}$)	$\sqrt{5} \approx 682/305$
$e = 2.71828183\dots$ continued fraction $[2; 1, 2, 1, 1, 4, 1, 1, 6, \dots]$ (pattern, not periodic) convergents: 2, 3, 8/3, 11/4, 19/7, 87/32, 106/39, 193/71 $87/32 = 2.71875\dots$ (error $\approx +5 \times 10^{-4}$) $193/71 = 2.7183099\dots$ (error $\approx +3 \times 10^{-5}$)	$e \approx 193/71$

The famous $22/7 \approx \pi$ works the same way — it is the convergent $[3; 7]$ of $\pi = [3; 7, 15, 1, 292, \dots]$.

Figure 1.1: For each of $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$, e : its continued-fraction expansion (periodic for the square roots, patterned for e) and the list of convergents with decimal values and errors. Recommended picks: $\sqrt{2} \approx 577/408$ (error 2×10^{-6}), $\sqrt{3} \approx 97/56$ (9×10^{-5}), $\sqrt{5} \approx 682/305$ (2×10^{-6}), $e \approx 193/71$ (3×10^{-5}). The convergents alternate over and under the true value and shrink in error rapidly. Note $22/7 \approx \pi$ is the convergent $[3; 7]$ of π .

1.6 Remark — Why Convergents Are the *Best* Approximations

A convergent p/q of a number x is not merely a good fraction — it is optimal in a precise sense: no other fraction with denominator $\leq q$ comes closer to x . Moreover it satisfies

$$\left| x - \frac{p}{q} \right| < \frac{1}{q^2},$$

so the error shrinks *faster* than the denominator grows. That is why **577/408** pins down $\sqrt{2}$ to six decimal places with a two-digit-ish denominator, while a “random” fraction like $1414/1000$ (bigger denominator!) is far worse. This $1/q^2$ quality is the whole point of the “rational approximation” field the problem alludes to.

1.7 Remark — The Convergents Bracket the Target

Reading down any convergent list, the values alternate: one is *too big*, the next *too small*, closing in on the true value like a shrinking interval. For $\sqrt{2}$: $1 < \frac{7}{5} < \frac{41}{29} < \sqrt{2} < \frac{99}{70} < \frac{17}{12} < \frac{3}{2}$. Even convergents approach from below, odd ones from above, and the gap collapses at each step. This bracketing is a free by-product of the continued-fraction machinery and gives, at every stage, both an over- and an under-estimate.

1.8 Remark — Periodic CFs and Quadratic Irrationals

Notice the square roots have *periodic* continued fractions ($\sqrt{2} = [1; \overline{2}]$, $\sqrt{5} = [2; \overline{4}]$, $\sqrt{3} = [1; \overline{1, 2}]$) while e does not. This is a theorem of Lagrange: a continued fraction is eventually periodic **iff** the number is a *quadratic irrational* — a root of a quadratic with integer coefficients, exactly what $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$ are. The periodicity is why the square-root approximations are so easy to extend: just repeat the block. The number e , being *transcendental* (not the root of any polynomial), cannot be periodic — yet it still has the beautifully regular pattern $[2; 1, 2, 1, 1, 4, 1, 1, 6, \dots]$, which is how Euler first proved e irrational.

1.9 Remark — A Shortcut for Square Roots (Newton's Method)

Continued fractions are the *optimal* tool, but there is a quick hands-on alternative for the square roots: Newton's iteration $x \mapsto \frac{1}{2}\left(x + \frac{N}{x}\right)$, which *doubles* the number of correct digits each step. Starting from $x = \frac{3}{2}$ for $\sqrt{2}$:

$$\frac{3}{2} \longmapsto \frac{1}{2}\left(\frac{3}{2} + \frac{2}{3/2}\right) = \frac{17}{12} \longmapsto \frac{1}{2}\left(\frac{17}{12} + \frac{2}{17/12}\right) = \frac{577}{408}.$$

Strikingly, this lands on the very same $\frac{17}{12}$ and $\frac{577}{408}$ that appear among the convergents — Newton's method here leaps directly to every *other* convergent. It is the practical way to get $\sqrt{N}$ to many digits by hand, and it explains why $577/408$ is such a natural answer for $\sqrt{2}$.

1.10 Remark — These Approximations Are Genuinely Old and Useful

$22/7$ for π and $17/12$ for $\sqrt{2}$ are ancient — the Babylonians used sexagesimal equivalents of such fractions for $\sqrt{2}$ four thousand years ago, and $22/7$ was known to Archimedes. The point of the exercise, sitting where it does right after the definition of the rationals, is conceptual: the irrationals $\sqrt{2}, \sqrt{3}, \sqrt{5}, e$ can never be written *exactly* as a fraction (that is what “irrational” means, and $\sqrt{2}$ ’s irrationality is proved on the very next page), yet they can be *pinned arbitrarily close* by fractions. The rationals are **dense** in the reals — you can get as near as you like — even though no rational ever hits an irrational on the nose.